

Project Executable Files

Date	3 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Project Executable Files

This section documents all executable and deployable files for the OptiCrop project. All files are properly organized, named, and available on GitHub. The evaluator can run the solution using the provided instructions below.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	NA — CSV dataset used
5	Environment configuration file (.env.example or similar)	NA — No API keys required
6	Deployed application URL (if hosted)	NA — Local deployment only
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA — Web application
8	Dockerfile / Containerization config (if applicable)	NA — Flask dev server used
9	Test files and test results	Yes — Performance Testing PDF
10	Demo video or walkthrough (if required)	Yes — Demo video uploaded to Drive

Step 2: File / Folder Structure

AI-ML-Track-Project-Template/ ■■■ 1. Brainstorming & Ideation/ ■■■ Brainstorming & Idea Prioritization.pdf ■■■ Define Problem Statements.pdf ■■■ Empathy Map.pdf ■■■ ER Diagram.pdf ■■■ 2. Requirement Analysis/ ■■■ Customer Journey Map.pdf ■■■ Data Flow Diagram.pdf ■■■ Solution Requirements.pdf ■■■ Technology Stack.pdf ■■■ 3. Project Design Phase/ ■■■ Problem-Solution Fit.pdf ■■■ Proposed Solution.pdf ■■■ Solution Architecture.pdf ■■■ 4. Project Planning Phase/ ■■■ Project Planning.pdf ■■■ 5. Project Development Phase/ ■■■ OptiCrop.ipynb ■■■ app.py ■■■ model.pkl ■■■ Crop_recommendation.csv ■■■ Coding & Solution.pdf ■■■ templates/ ■■■ home.html ■■■ about.html ■■■ findyourcrop.html ■■■ 6. Project Testing/ ■■■ Performance Testing.pdf ■■■ 7. Project Documentation/ ■■■ Project Executable Files.pdf ■■■ Sample Project Documentation.pdf ■■■ 8. Project Demonstration/ ■■■ [Demonstration PDFs] ■■■ README.md

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Local — http://127.0.0.1:5000 (Flask Development Server)
Login Credentials (Demo)	No authentication required — open access web application

Platform / Hosting Provider	Local Machine — Flask built-in development server
Repository Link	https://github.com/hemanth5267/AI-ML-and-GEN-AI-Track-Project-Template
Demo Video Link	Google Drive link — to be uploaded after recording demo video

Step 4: Run Instructions

Pre-requisites: • Python 3.10+, pip, Git installed • Libraries: flask, scikit-learn, pandas, numpy, matplotlib, seaborn, jupyter Step 1: Clone the repository `git clone https://github.com/hemanth5267/AI-ML-and-GEN-AI-Track-Project-Template` Step 2: Navigate to development folder `cd "AI-ML-and-GEN-AI-Track-Project-Template/5. Project Development Phase"` Step 3: Create and activate virtual environment `python3 -m venv source venv/bin/activate` (Mac/Linux) Step 4: Install dependencies `pip install flask scikit-learn pandas numpy matplotlib seaborn jupyter` Step 5: Run Jupyter Notebook to train and save model `jupyter notebook OptiCrop.ipynb` → Run all cells → `model.pkl` will be generated Step 6: Run Flask application `python3 app.py` Step 7: Open browser <http://127.0.0.1:5000>

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Application runs only on local server — not cloud deployed	Can be deployed on Heroku/Render in future — Resolved for local demo

2	Logistic Regression shows convergence warning for low max_iter	Fixed by setting max_iter=1000 in model training
3	No user authentication implemented in current version	Planned for future version — not required for current scope